JUTC Bus Tracking System User Manual

This manual explains how passengers, drivers and administrators use the website. The system is an academic demonstration using simulated bus locations and arrival estimates; it does not show official live JUTC operations.

Passengers

Open the website address supplied by the project group. Passengers do not need an account.

Find a route

1. Choose a **Starting stop** and **Destination** from the available suggestions.
2. Select **Find a route**. The results show matching direct routes and the stops along your journey.
3. Use **Search routes** to filter the route list by a route or stop name. If no direct journey is found, try another pair of stops.

The website currently searches direct journeys; it does not plan transfers between buses.

View buses and arrival estimates

The map shows sample routes, stops and individually labelled buses. The bus list identifies each bus, its service state and next stop.

In **Estimated arrivals**, select a route, a specific bus and a bus stop. The estimate uses the saved simulated position, average speed and any published delay minutes. Bus information refreshes about every five seconds.

A parked bus has no arrival estimate. If location updates stop for more than one minute, the bus is marked stale and its estimate becomes unavailable.

Receive alerts

Select **Watch this bus** after choosing a bus and stop. An alert appears when the running bus is within 1 km of that stop or has a reported delay. Select **Stop watching** to cancel; changing your selection also cancels the watch.

Use **Enable device notifications** if you want additional browser notifications, then allow permission. Keep the page open. In-page alerts still work when device notifications are unavailable. SMS and closed-browser notifications are not provided.

The **Service information** section displays administrator announcements about delays, route changes and service disruptions.

Driver and Administrator Instructions

Drivers

1. Open **Driver console** and sign in with an authorised app account. Select one of your assigned buses.
2. Select **Start engine**. Keep the driver page open so simulated positions can be saved about every five seconds.
3. Choose **On time** or **Delayed** and select **Update status** when the service state changes.
4. Select **Stop engine** to freeze the saved position, then sign out when finished.

Drivers can operate only their assigned buses. An administrator can operate any bus for demonstrations. If no buses appear, ask an administrator to assign one. Closing or backgrounding the page can interrupt updates.

Administrators

Open **Route management** and sign in with an administrator account. The configured shared demo account accepts **JUTCAdmin2026** as its username. Use the password set by the project operator.

Routes: Enter a route name and at least two existing stops, separated by commas in travel order. Use the edit control to change a route and save it. Saving an edited route parks its buses; restart them from the driver console afterward.

Delete: Confirm only when you intend to remove a route. Deletion also removes its buses, driver assignments and route-specific alerts.

Stops and buses: Add a stop using a unique name and valid coordinates. Add a bus using a unique label and an available route. New buses start parked.

Driver assignments: Select a parked bus and enter the driver's Supabase Auth user ID. The account must have a driver role. Leave the field empty to unassign the bus.

Service alerts: Select an alert type and route, enter a message and delay minutes where applicable, then select **Publish alert**. Use whole delay minutes from 0 to 120. Remove an alert when it no longer applies.

Common problems

Login or role error: Ask the operator to check the app account, password and assigned role. Supabase project membership alone is not an app login. **Unavailable or stale arrivals:** Check that the driver page is open, the engine is on and the connection works.

Project sources: Pseudocode Description, sections 1 to 3; Software Implementation Group Work, pages 5 to 6 and 10 to 12. The report permits simulated tracking.